

CSCI 5541 NLP HW4

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1 Introduction and Set up

The assignment is to build a complete text generation research pipeline using existing tools and pre-trained models. The pipeline will cover decoding algorithms, automatic evaluation, human evaluation, and analysis of outputs. The structure of this article is the same as the instruction of the assignment.

2 Intrinsic Evaluation

2.1 Select a Model and Write Prompts

I choose model Qwen3-0.6B in this part. And prompts:

1. "Once upon a time"
2. "The future of artificial intelligence"
3. "In a Taiwan"
4. "The best way to learn"
5. "To be or not to be"

2.2 Implement Decoding Algorithms

First, I set max new tokens at 80 in each algorithm.

In Greedy decoding, I set do sample as False, which will let algorithm choose the token with largest probability.

In Beam search set do sample as True, num beans at 5 early stopping at true.

In Top-k sampling, I set do sample as True, top k as 50.

In Top-p sampling, I set do sample as True, top p as 0.95.

Finally, I show the average perplexity of each algorithm at following:

Index	Decoding Method	Average Perplexity
0	greedy	3.8481
1	beam	2.4586
2	top_k	4.388
3	top_p	5.002

In summary, above table comparing different algorithm, the perplexity data faithfully reflects the design essence of the algorithms: Beam Search achieves the lowest mathematical perplexity (2.4586) through global search; while Top-p and Top-k deliberately select words with non-absolutely high probabilities in pursuit of text diversity and creativity, thus exhibiting higher perplexity (5.002 and 4.388, respectively).

3 Extrinsic Evaluation

3.1 Select a Task and Decode Samples

I select Option 2 and implement 4 algorithm in this part.

Task: Summarization

Model: T5-Finetuned-Summarization-DialogueDatasetLinks

Dataset: knkarthick/dialogsumLinks

3.2 Automatic Evaluation

I choose rouge as content-overlap metric and bertscore as the model based metric(RougeL and F1 in detail). The average metric score for each decoding algorithm across all 50 samples and compare the two metrics are following:

Index	Method	Overlap Mean	Model Mean
0	greedy	0.2936	0.8343
1	beam	0.2849	0.8402
2	top_k	0.2456	0.8269
3	top_p	0.2621	0.8272

Table 1: Comparison of decoding methods by overlap mean and model mean.

I extracted the ROUGE-L score to measure structural coherence and the BERTScore F1 score to achieve a balanced semantic evaluation. Specifically, ROUGE-L calculates the Longest Common Subsequence, which captures word order and

sentence-level fluency better than strict n-gram matching, effectively penalizing outputs that lack grammatical structure. Meanwhile, BERTScore leverages contextual embeddings to evaluate semantic similarity, accommodating valid paraphrasing. By extracting its F1 score—the harmonic mean of Precision and Recall—I can holistically penalize both model hallucinations (via Precision) and missing reference information (via Recall), ensuring the evaluation is fair and not biased by excessively long or short generated sequences.

3.3 Human Evaluation

Following table is the result after I evaluation. After I check, some of them even not can be classify as a sentence. And there exist illusion in others responses. Specific top k and top p algorithm can generate a fluency sentences, but neither of them can be summery the article correctly.

Index	Method	Overlap Mean	Model Mean	Factuality Mean	Fluency Mean	Coherence Mean
0	greedy	0.2936	0.8343	2.4	1.8	1.8
1	beam	0.2849	0.8402	1.6	2.2	1.8
2	top_k	0.2456	0.8269	1.4	3.0	1.6
3	top_p	0.2621	0.8272	1.6	2.0	1.8

Table 2: Comparison of decoding methods on overlap, model score, factuality, fluency, and coherence.

In NLP, no single metric is perfect. While automatic metrics can effectively measure whether a model has grasped the key points of a standard answer, they cannot reflect the true human experience of 'natural language'. For deploying models in the real world to interact with users, the high fluency provided by Top-k/Top-p metrics, which rely on randomness, is often more critical, even if they don't perform well on automatic metrics.